EXPERIMENT NO: 05

Data Collection and Initial Exploration

Program:

```
import pandas as pd
diabetes_df=pd.read_csv("diabetes (1).csv")
print(diabetes_df.head())
print(diabetes_df.info())
print(diabetes_df.describe())
import matplotlib.pyplot as plt
import seaborn as sns
diabetes_df.hist(bins=50,figsize=(20,15))
plt.show()
sns.pairplot(diabetes_df)
plt.show()
```

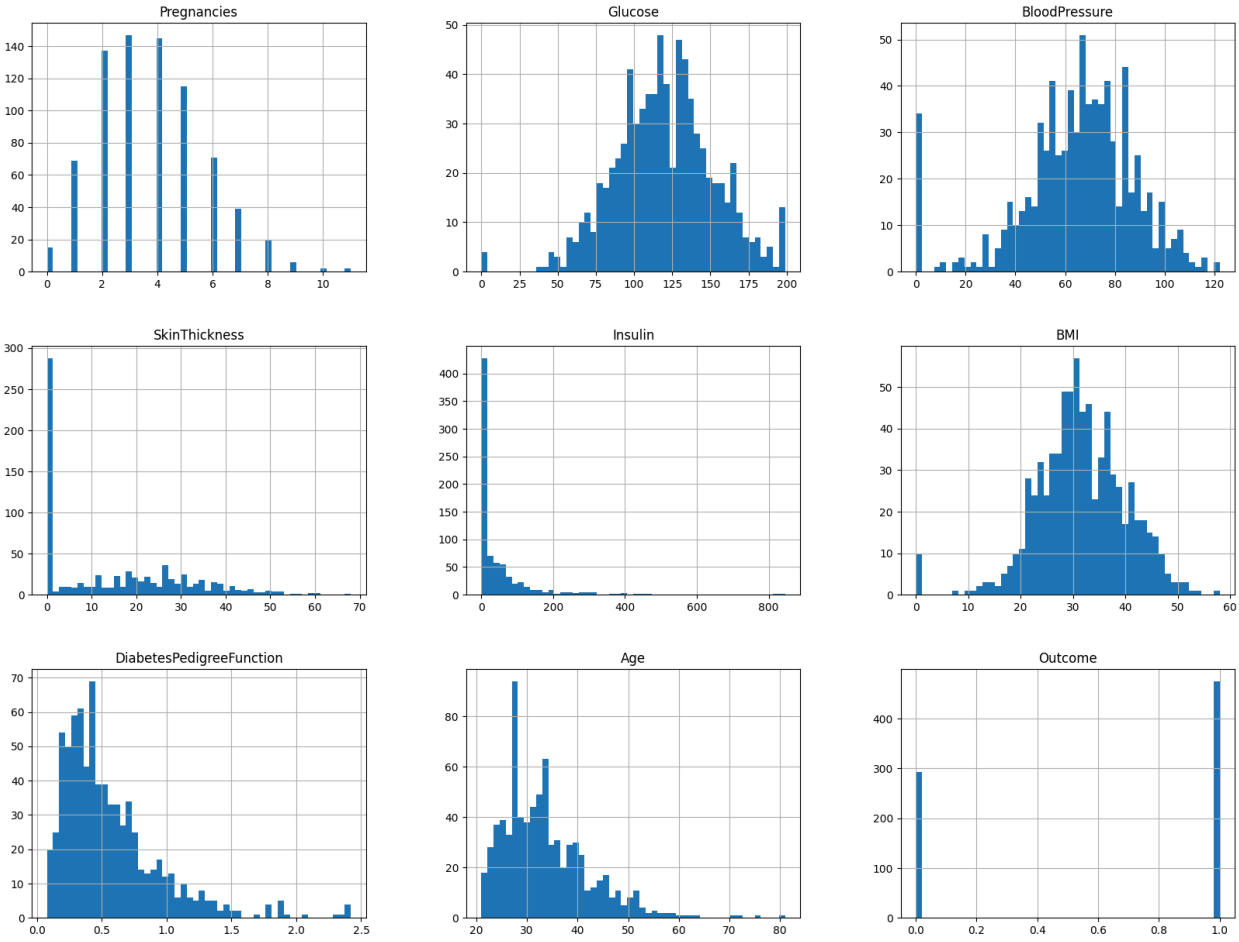
Output:

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
```

```
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.768229  120.165365      64.972656      15.522135   44.391927
std       1.946060   32.387525      23.193434      15.576602   84.220647
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       2.000000   98.000000      53.000000       0.000000    0.000000
50%       4.000000  119.000000      67.000000      13.000000    5.000000
75%       5.000000  140.000000      80.000000      27.000000   55.250000
max      11.000000  199.000000     122.000000      68.000000  846.000000

              BMI  DiabetesPedigreeFunction        Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.616276                  0.563865   33.558594    0.618490
std      8.581108                  0.384463    8.546778    0.486074
min      0.000000                  0.078000   21.000000    0.000000
25%     26.450000                  0.299750   27.000000    0.000000
50%     31.300000                  0.454500   32.000000    1.000000
75%     37.100000                  0.717000   38.000000    1.000000
max     58.000000                  2.420000   81.000000    1.000000
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome


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